NN-EUCLID: deep-learning hyperelasticity without stress data

Prakash Thakolkaran[a], Akshay Joshi[a], Yiwen Zheng[a], Moritz Flaschel[b], Laura De Lorenzis[b], Siddhant Kumar[a,*]

[a] *Department of Materials Science and Engineering, Delft University of Technology, 2628 CD Delft, The Netherlands*

[b] *Department of Mechanical and Process Engineering, ETH Zürich, 8092 Zürich, Switzerland*

Corrections to the original article

The original article (Thakolkaran et al., 2022) published by the authors contain the following typographical mistake.

Equation 28 in the original article:

$$W(\boldsymbol{F}) = \frac{\mu}{\eta}(\lambda_1^\eta + \lambda_2^\eta + \lambda_3^\eta - 3)$$

should be corrected to:

$$W(\boldsymbol{F}) = \frac{\mu}{\eta}\left(\tilde{\lambda}_1^\eta + \tilde{\lambda}_2^\eta + \tilde{\lambda}_3^\eta - 3\right) + 1.5(J-1)^2 \qquad \text{with} \qquad \tilde{\lambda}_k = J^{-1/3}\lambda_k,\ k = 1, 2, 3,$$

Additionally, the parity plot in Figure 10 of the original article contains a typo in the axis labels: $\tilde{I}_1$ and J should be $(\tilde{I}_1 - 3)$ and $(J - 1)$, respectively. Please refer to Figure 1 for the corrected section.

References

Thakolkaran, P., Joshi, A., Zheng, Y., Flaschel, M., De Lorenzis, L., Kumar, S., 2022. NN-EUCLID: Deep-learning hyperelasticity without stress data. Journal of the Mechanics and Physics of Solids 169, 105076. URL: http://dx.doi.org/10.1016/j.jmps.2022.105076, doi:10.1016/j.jmps.2022.105076.

*Email: Sid.Kumar@tudelft.nl

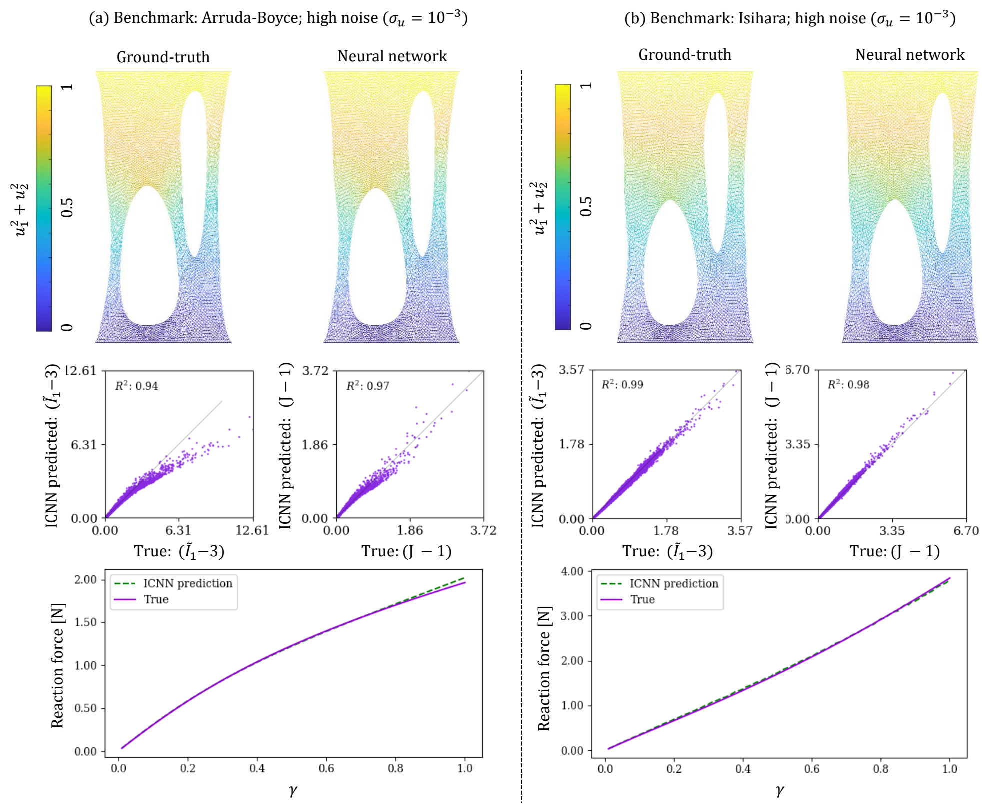

Figure 1: Comparison of FEM solutions for the validation specimen (Figure 3b) between the ground-truth and ICNN-based constitutive models for (a) Arruda-Boyce (27) and (b) Isihara (24) benchmarks. (*Top*) Deformed shape (shaded by displacement magnitude) of the specimen at loading parameter $\delta = 1$ (see Figure 3b for details). (*Middle*) Predicted strain invariants ($(\tilde{I}_1 - 3)$ and $(J - 1)$) at each quadrature point from the ICNN-based simulation vs. true strain invariants (from ground-truth simulation) in the validation specimen. In the ideal case, we expect each data point to lie on a line (indicated as red dashed line) with zero intercept and unit slope for perfect accuracy. The coefficient of determination (R^2) with respect to this line serves as a measure of accuracy. (*Bottom*) Comparison of the vertical reaction force on the top surface as a function of loading parameter δ, obtained from the ICNN-based simulation vs. the ground-truth simulation.